

UTM TENSILE REPORT — UTM_Test_20260818_111324

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

46.3 MPa

UTS

43.3 MPa

Yield σ_y

2.86 GPa

Modulus E

5.2 %

Failure ϵ_f

2015 kJ/m³

Toughness

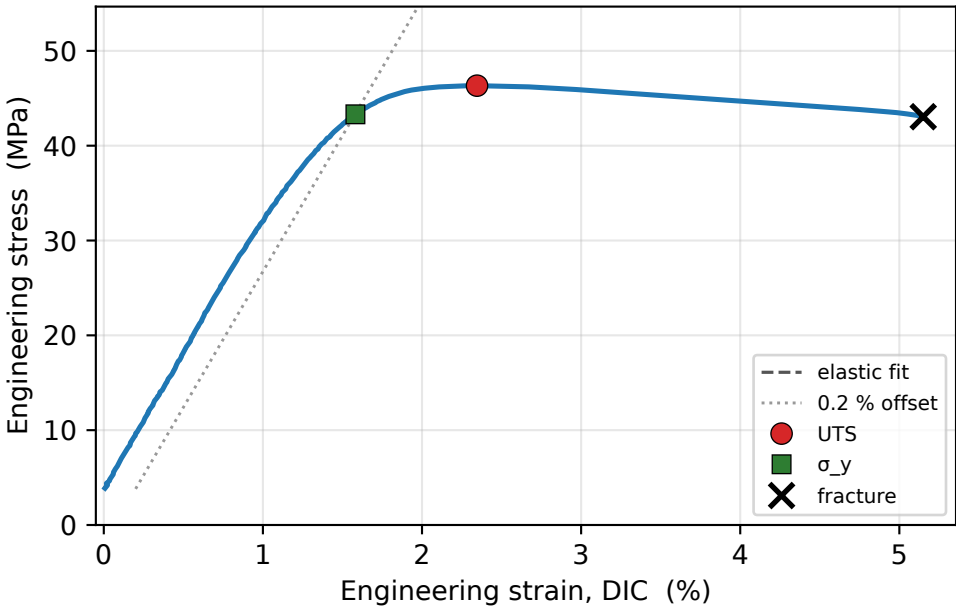
307 N

Preload anchor

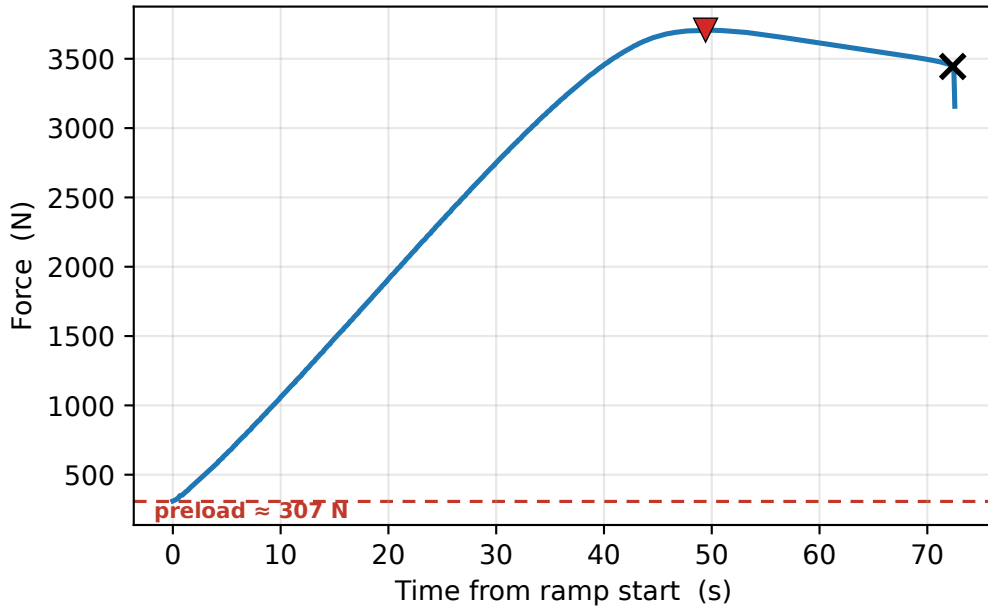
TEST SETTINGS

Specimen	UTM_Test_20260818_111324
DIC mode	Black
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	100 %
Force calib	sc -0.0065 / off 331.87
Comment	—
Test date	2026-08-18 11:11:26
Duration	123.5 s

Stress-strain



Load vs time

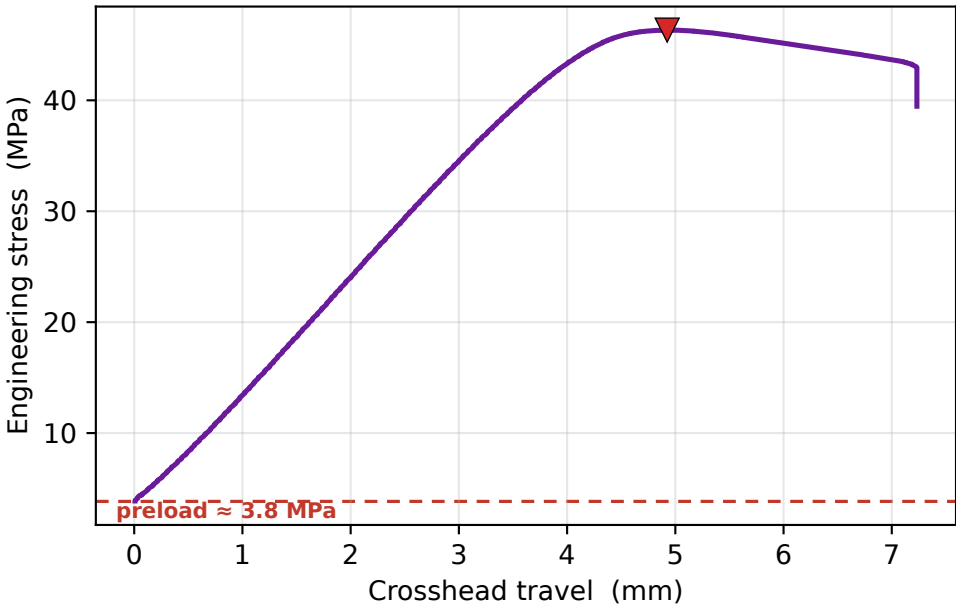


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	46.3	32-60 PASS	×1.25
σ_y	43.3	30-50 PASS	×1.34
E	2.9	3-5.5 low	×1.00
ϵ_f	5.2 %	(range 3-7 %)	×1.55

✓ strength inside journal range ($k \approx 1$)

Stress vs crosshead



DIC strain vs crosshead — the compliance gap

